1D 1 EMPIRICAL FORMULAE

SPECIFICATION REFERENCE 1.1 1.6

LEARNING OBJECTIVES

- Know the term empirical formula.
- Use experimental data to calculate empirical formulae.

INTRODUCTION TO EMPIRICAL FORMULAE

The three letters 'mol' appear in several words used in chemistry. You will know some of these from your previous studies, especially 'molecule', which is a group of two or more atoms joined together by covalent bonds. You know that a molecule of water can be represented by the formula H_2O, so this is the molecular formula of water. You have already come across the term 'relative molecular mass', which you will probably remember is 18 (or more accurately, 18.0) for water.

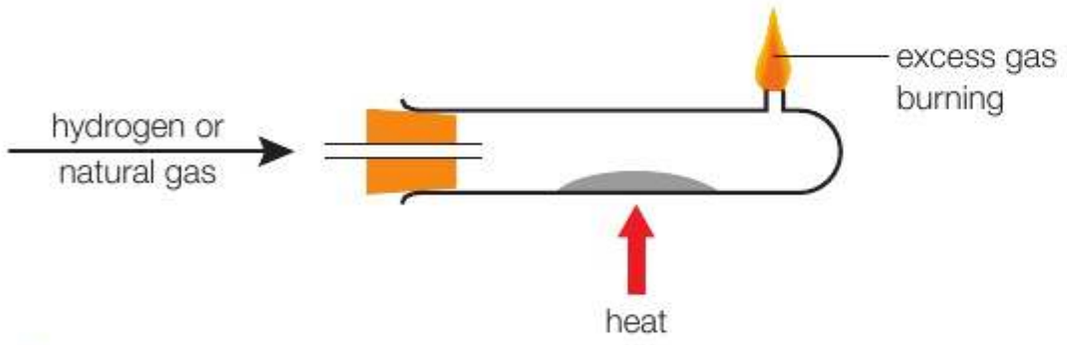
The term 'empirical' indicates that some information has been found by experiment. An **empirical formula** shows the smallest whole number ratio of the atoms of each element in a compound.

ONE EXAMPLE OF AN EXPERIMENTAL METHOD

A simple example involves determining the formula of an oxide of copper. This oxide can be converted to copper by removing the oxygen.

Here are the steps in the experiment.

- Place a known mass of the oxide of copper in the tube.
- Heat the oxide in a stream of hydrogen gas (or natural gas, which is mostly methane).
- The gas reacts with the oxygen in the copper oxide and forms steam.
- The colour of the solid gradually changes to orange-brown, which is the colour of copper.
- The excess gas is burned off at the end of the tube for safety reasons.
- After cooling, remove and weigh the solid copper.
- It is good practice to heat the solid again in the stream of the gas to check whether its mass changes. Heating to a constant mass suggests that the conversion to copper is complete.



▲ **fig A** This apparatus can be used to convert copper oxide to copper by removing the oxygen.

CALCULATING EMPIRICAL FORMULAE

The calculation method involves these steps.

- Divide the mass, or percentage composition by mass, of each element by its relative atomic mass.
- If necessary, divide the answers from this step by the smallest of the numbers.
- This gives numbers that should be in an obvious whole number ratio, such as 1 : 2 or 3 : 2.
- These whole numbers are used to write the empirical formula.

The numbers may not be in an exact ratio because of experimental error, but you should be able to decide what the nearest whole-number ratio is.

Use at least two significant figures in the calculation (preferably three), and beware of inappropriate rounding. For example, you cannot convert a 1.2 : 1.2 : 1.9 ratio to 1 : 1 : 2, because that would give you the wrong answers.

It may help you to organise the calculation using a table, although this is not essential.

CALCULATION USING MASSES

Assume that these are the results of the experiment outlined on the left.

mass of copper oxide = 4.28 g

mass of copper = 3.43 g

mass of oxygen removed is 4.28 − 3.43 = 0.85 g

Table A is the calculation table for these results.

	Cu	O
mass of element / g	3.43	0.85
relative atomic mass	63.5	16.0
division by A_r	0.0540	0.0531
ratio	1	1

table A

Here, the ratio is obviously 1 : 1, so the empirical formula is CuO.

CALCULATION USING PERCENTAGE COMPOSITION BY MASS

You do the calculation in the same way, except that you divide percentages instead of masses by the relative atomic masses.

The calculation table, **table B**, refers to a compound containing three elements. The compound has the percentage composition by mass C = 38.4%, H = 4.8%, Cl = 56.8%.

	C	H	Cl
% of element	38.4	4.8	56.8
relative atomic mass	12.0	1.0	35.5
division by A_r	3.2	4.8	1.6
ratio	2	3	1

table B

You can see that the empirical formula is C_2H_3Cl.

CALCULATION WHEN THE OXYGEN VALUE IS NOT PROVIDED

The results for some compounds do not include values for oxygen because it is often difficult to obtain an experimental value for the mass of oxygen. Sometimes you will need to remember to calculate the percentage of oxygen by subtraction. Here is an example.

A compound has the percentage composition by mass Na = 29.1%, S = 40.5%, with the remainder being oxygen.

The percentage of oxygen = 100 − (29.1 + 40.5) = 30.4%.

Table C is the calculation table.

	Na	S	O
% of element	29.1	40.5	30.4
relative atomic mass	23.0	32.1	16.0
division by A_r	1.27	1.26	1.90
division by the smallest	1	1	1.5
ratio	2	2	3

table C

You can now see that the empirical formula is $Na_2S_2O_3$.

EXAM HINT

Take care to look out for decimals that indicate obvious fractions, e.g., simplest ratio 1 : 1.33 indicates an empirical formula of 3 : 4. How would a ratio of 1 : 1.25 convert into an empirical formula?

CALCULATION USING COMBUSTION ANALYSIS

Many organic compounds contain carbon and hydrogen, or carbon, hydrogen and oxygen. When a known mass of an organic compound is completely burned, it is possible to collect and measure the masses of carbon dioxide and water formed. The calculation is more complex because there are extra steps. Here is an example.

A 1.87 g sample of an organic compound was completely burned, forming 2.65 g of carbon dioxide and 1.63 g of water.

In this type of calculation, the first steps are to calculate the masses of carbon and hydrogen in the carbon dioxide and water.

- The relative molecular mass of carbon dioxide is 44.0 but, because the relative atomic mass of carbon is 12.0, the proportion of carbon in carbon dioxide is always 12.0 ÷ 44.0.
- Similarly, the proportion of hydrogen in water is always (2 × 1.0) ÷ 18.0.

All of the carbon in the carbon dioxide comes from the carbon in the organic compound. Similarly, all of the hydrogen in the water comes from the hydrogen in the organic compound. In this example:

$$\text{mass of carbon} = \frac{2.65 \times 12.0}{44.0} = 0.723\,\text{g}$$

$$\text{mass of hydrogen} = \frac{1.63 \times 2.0}{18.0} = 0.181\,\text{g}$$

These two masses add up to 0.904 g.

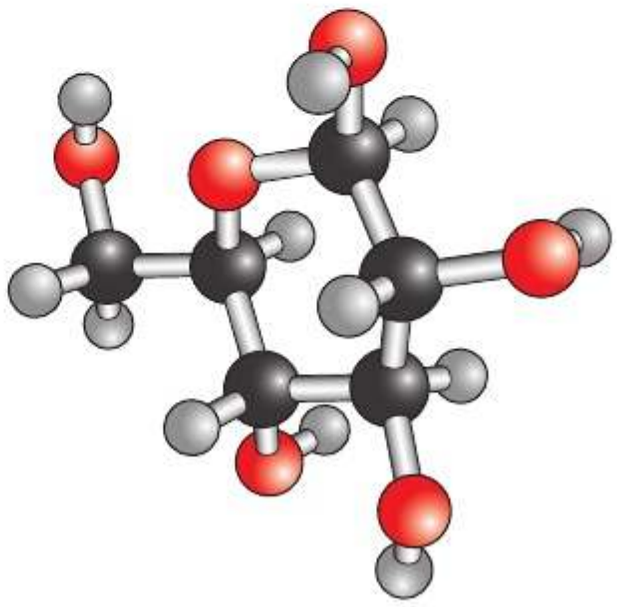
The original mass of the organic compound was 1.87 g, so the difference must be the mass of oxygen present in the organic compound. The mass of oxygen = 1.87 − 0.904 = 0.966 g.

Table D is the calculation table. You can see that the empirical formula of the sample compound is CH_3O.

	C	H	O
mass of element / g	0.723	0.181	0.966
relative atomic mass	12.0	1.0	16.0
division by A_r	0.0603	0.181	0.0604
ratio	1	3	1

table D

Fig B shows a model of the glucose molecule. You can count the numbers of the three different atoms in the molecule.



▲ **fig B** A ball-and-stick model of glucose.

LEARNING TIP

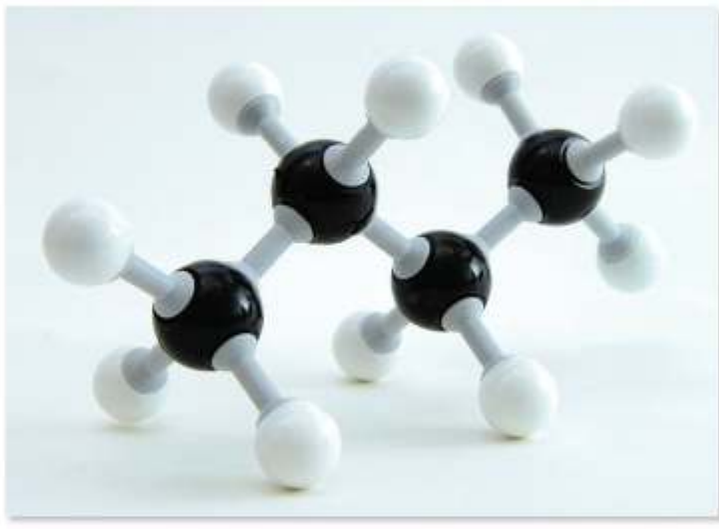
Divide by the relative atomic mass, not the atomic number or the relative molecular mass.

For oxygen, only divide by 16.0, not by 8 or 32.

CHECKPOINT

1. A compound has the percentage composition by mass Ca = 24.4%, N = 17.1% and O = 58.5%.
What is its empirical formula?

2. Combustion analysis of 2.16 g of an organic compound produced 4.33 g of carbon dioxide and 1.77 g of water.
What is its empirical formula?



3. Can you work out the empirical formula of this molecule? Black represents carbon and white represents hydrogen.

SUBJECT VOCABULARY

empirical formula the smallest whole-number ratio of atoms of each element in a compound

1D 2 MOLECULAR FORMULAE

SPECIFICATION REFERENCE
1.1 1.6 1.8(iii)

LEARNING OBJECTIVES

- Know the term molecular formula.
- Use experimental data to calculate molecular formulae.
- Use the expression $pV = nRT$ for gases and volatile liquids.

INTRODUCTION TO MOLECULAR FORMULAE

In **Topic 1D.1**, you learned how to calculate empirical formulae. Sometimes the empirical formula of a compound is the same as its molecular formula. Carbon dioxide (CO_2) and water (H_2O) are common examples you will know. The **molecular formula** of a compound shows the actual numbers of the atoms of each element in the compound.

Examples of compounds with different empirical and molecular formulae are:

hydrogen peroxide	empirical formula is HO	molecular formula is H_2O_2
butane	empirical formula is C_2H_5	molecular formula is C_4H_{10}

To determine the molecular formula of a compound, you need to already know or to calculate:
- the empirical formula
- the relative formula mass.

For example, if you had already found that the empirical formula of a compound was HO and you then found that its relative formula mass was 34, you could compare the relative mass of the empirical formula (17) with 34 and work out that the molecular formula was double the empirical formula.

CALCULATING MOLECULAR FORMULAE

You have already practised calculating an empirical formula from experimental data. Now look at how to calculate a molecular formula from an empirical formula.

WORKED EXAMPLE 1

In this example, the empirical formula is given.
A compound has the empirical formula CH and a relative formula mass of 104.

The 'formula mass' of the empirical formula is 13.0.

104 ÷ 13.0 = 8, so the molecular formula of the compound is C_8H_8.

WORKED EXAMPLE 2

In this example, you first have to work out the empirical formula. A compound contains the percentage composition by mass Na = 34.3%, C = 17.9%, O = 47.8%, and has a **molar mass** of 134 g mol^{-1}.
The calculations are shown below.

	Na	C	O
% of element	34.3	17.9	47.8
relative atomic mass	23.0	12.0	16.0
division by A_r	1.49	1.49	2.99
ratio	1	1	2

The empirical formula is $NaCO_2$.

The 'formula mass' of the empirical formula is 23.0 + 12.0 + (2 × 16.0) = 67.0.

The molar mass, 134, is 2 × 67.0, so the molecular formula of the compound is $Na_2C_2O_4$.

THE IDEAL GAS EQUATION $pV = nRT$

$pV = nRT$ is the ideal gas equation, and can be used for gases (or volatile liquids above their boiling temperatures) to find the amount of a substance in moles. If the mass of the substance is also known, then the molar mass of the substance can be calculated. This gives the extra information needed to work out a molecular formula from an empirical formula.

The expression can also be rearranged to calculate a value of p, or V or T.

SI UNITS

When using this equation, you need to be careful that the units are the correct ones. It is always safest to work in SI units.

The SI units you should use are:

- p = pressure in pascals (Pa)
- V = volume in cubic metres (m^3)
- T = temperature in kelvin (K)
- n = amount of substance in moles (mol)
- R = the gas constant – this appears in the Data Booklet provided for use in the examinations and has the value $8.31\,J\,mol^{-1}K^{-1}$.

Sometimes in a question you may find that the units quoted are not SI units. If this is the case, then you will need to convert them to SI units. **Table A** shows the conversions you are likely to need.

CONVERSION	HOW TO DO IT
kPa → Pa	multiply by 10^3
$cm^3 \rightarrow m^3$	divide by 10^6 or multiply by 10^{-6}
$dm^3 \rightarrow m^3$	divide by 10^3 or multiply by 10^{-3}
°C → K	add 273

table A

WORKED EXAMPLE 3

In this example, you will calculate the molar mass of the gas. It may help you (at least until you have had more practice) to write a list of the values with any necessary conversions.

A 0.280 g sample of a gas has a volume of 58.5 cm^3, measured at a pressure of 120 kPa and a temperature of 70 °C. Calculate the molar mass of the gas.

$p = 120\,kPa = 120 \times 10^3\,Pa$

$V = 58.5\,cm^3 = 58.5 \times 10^{-6}\,m^3$

$T = 70\,°C = 343\,K$

$R = 8.31\,J\,mol^{-1}K^{-1}$

So, $n = \frac{pV}{RT} = \frac{120 \times 10^3 \times 58.5 \times 10^{-6}}{8.31 \times 343} = 0.00246\,mol$

$M = \frac{m}{n} = \frac{0.280}{0.00246} = 114\,g\,mol^{-1}$

WORKED EXAMPLE 4

In this example, you will calculate the empirical formula, then the amount in moles. After that the molar mass, then the molecular formula.

A compound has the percentage composition by mass C = 52.2%, H = 13.0%, O = 34.8%. A sample containing 0.173 g of the compound had a volume of 95.0 cm^3 when measured at 105 kPa and 45 °C. What is the molecular formula of this compound?

Step 1: calculate the empirical formula

	C	H	O
% by mass of element	52.2	13.0	34.8
relative atomic mass	12.0	1.0	16.0
division by A_r	4.35	13.0	2.175
ratio	2	6	1

The empirical formula is C_2H_6O.

Step 2: calculate the amount in moles

$n = \frac{pV}{RT} = \frac{105 \times 10^3 \times 95.0 \times 10^{-6}}{8.31 \times 318} = 0.00377\,mol$

Step 3: calculate the molar mass

$M = \frac{m}{n} = \frac{0.173}{0.00377} = 45.9\,g\,mol^{-1}$

Step 4: calculate the molecular formula

The 'formula mass' of the empirical formula is

$(2 \times 12.0) + (6 \times 1.0) + 16.0 = 46.0$

As 46.0 is the same as the molar mass, then the empirical and molecular formulae are the same.

The molecular formula is C_2H_6O.

LEARNING TIP

Be careful when using the word 'amount'. It should only be used for the amount, in moles, of a substance. For example, 'The amount of magnesium used was 0.150 mol' is correct.

You should not use it instead of quantities with other units. For example, 'The amount of magnesium used was 3.6 g' should be 'The mass of magnesium used was 3.6 g'.

Another example:

'The amount of water used was 25.0 cm^3' should be 'The volume of water used was 25.0 cm^3'.

CHECKPOINT

SKILLS PROBLEM SOLVING

1. A 2.82 g sample of a gas has a volume of 1.26 dm^3, measured at a pressure of 103 kPa and a temperature of 55 °C. Calculate the molar mass of the gas.
2. A compound has the percentage composition by mass C = 40.0%, H = 6.7%, O = 53.3%. A sample containing 0.146 g of the compound had a volume of 69.5 cm^3 when measured at 98 kPa and 63 °C. What is the molecular formula of this compound?

SUBJECT VOCABULARY

molecular formula the actual number of atoms of each element in a molecule

molar mass the mass per mole of a substance; it has the symbol M and the units $g\,mol^{-1}$